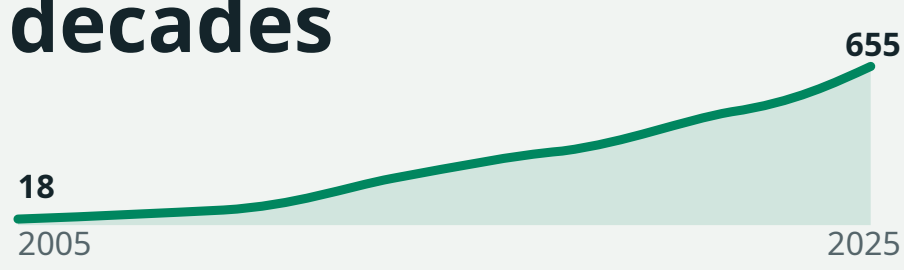


Perturbation evidence is multiplying faster than it can be organised

A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. The Perturbation Catalogue aims to unify datasets across modalities and provide a shared access path for this evidence.

36× growth in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: [FirstWord Pharma story 7139212](#).

Our approach turns scattered datasets into a unified resource

1,237
datasets

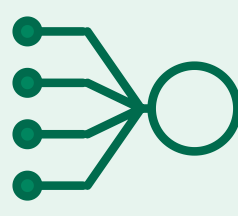
19,235
targets

36
tissues

34
cell types

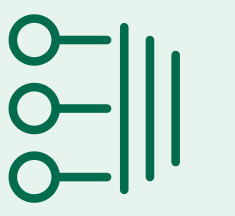
1,200
cell lines

169
diseases



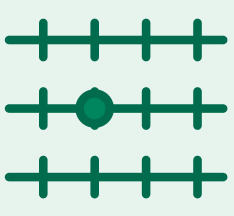
CRISPR screens · 1,201 datasets

Pooled guides → gene-level phenotypes and fitness.



Perturb-Seq · 26 datasets

Guide identity + scRNA-seq → cell-state responses.



MAVE · 10 datasets

Variant libraries + assays → sequence-function maps.

We extensively curate metadata...

84
meta-
data
fields

Study & provenance

Title, authors, year, URI, licence

Model system

Organism, tissue, cell line, disease

Perturbation & library

Targets, guides, library, variants

Readout & processing

Assay, technology, kit, reference

Stable **Ensembl gene IDs** + ontology terms make cross-study questions searchable.

And reprocess data from scratch

SRA / FASTQ

Raw sequencing reads and study-specific probe lists

→

Cell × gene counts

Shared reference and gene annotation

→

Guide assignment

Map cells to perturbations

→

QC + filtering

Produce the final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA. More datasets and modalities coming soon.

Browse the Catalogue

Search → Context → Filter → Explore

ebi.ac.uk/perturbation-catalogue

Perturbation Catalogue

Targets Datasets API Documentation About Request dataset

Perturbation Catalogue

Targets Search by metadata fields Search

Try searching for:

Perturbation Catalogue is a curated database that brings together data from various genetic perturbation experiments, including Perturb-Seq, CRISPR and MAVE screens, making it easier for researchers to study how modifying genes or proteins affects biological function across various biological and molecular contexts.

Perturbation Catalogue at a glance

Explore high-level statistics across the catalogue before diving into specific targets.

DATASETS
1237 (spanning 2014 - 2025)

TARGETS
19235

UNIQUE TISSUES
36

UNIQUE CELL TYPES
34

UNIQUE CELL LINES
1200

UNIQUE DISEASES
169

/targets?search=BRCA2

Browse Targets

Target	Library	Assay	Year	Cell Type	Condition
BRCA2	CRISPR	scRNA-seq	2025	HeLa	Control
BRCA2	CRISPR	scRNA-seq	2025	HeLa	BRCA2
BRCA2	CRISPR	scRNA-seq	2025	HeLa	BRCA2

/dataset/nadig_2025_jurkat

Dataset: Nadig 2025 Jurkat

Study Title: Transcriptional analysis of differential expression in perturbation assays

Lead Author: Alex Tsukanov

Study Year: 2025

Experiment Title: Jurkat Day 7 essential gene set Perturb-Seq experiment

Experiment Summary: Jurkat cells were transfected with a sgRNA library targeting a set of 1200 essential genes and analysed by single-cell RNA sequencing. Differential expression analysis was performed using DESeq2. The results show that the perturbation of essential genes leads to a significant decrease in gene expression.

Study Type: Perturb-Seq

Perturbation Type: CRISPR

Other Metadata Fields: ...

Data

Download Metadata Download Data

Access records via API

Structured JSON for custom pipelines, reproducible analysis and machine learning.

```
GET /v1/perturb-seq/nadig_2025_jurkat/search?
perturbation_gene_name=TFAM&limit=1
```

```
{
  "perturbation": { "gene_symbol": "TFAM" },
  "effect": { "gene_symbol": "MT-ND5", "log2fc": -1.926, "padj": 0.0 }
}
```

Download entire datasets

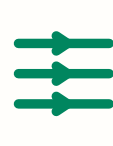
Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

Stay tuned for more exciting things coming soon



Extract structured metadata faster

AI-assisted drafting with curator review and validated ontology mapping.



Reanalyse more designs

More kits, experimental structures and independent quality metrics.



Publish regular, versioned releases

Explicit provenance and stable snapshots for reproducible downstream use.



Reanalyse MAVE data

Enrich variant-to-function coverage with harmonised scores.